DAY_01

DATE : 12.10.2026

TIME : 6:30 P.PM.TO 8:30 P.M.

1. Implement
    - Conditional Statements:
        I. If
        II. else if
        III. switch
    - Loops
        i. for loop
        ii. while loop
        iii. do while loop
    - Jump statement
        i. Break
        ii. continue
        iii. return keyword

DAY_02

DATE : 13.10.2026

TIME : 6:30 P.PM.TO 8:30 P.M.

1. Implement functions through your program.
2. Implement pointer through your program

DAY_03

DATE : 14.10.2026

TIME : 6:30 P.PM.TO 8:30 P.M.

1. Declare function and call it by reference and note the observations
2. Assignments using pointers, arrays of pointers.
3. Write a menu-driven C++ program to perform the following operations on a matrix of size (3 X 3).
    1. Accept matrix elements from the user.
    2. Display the matrix.
    3. Calculate the sum of all elements.
    4. Calculate the sum of each row.
    5. Calculate the sum of each column.
    6. Find the largest element.
    7. Find the smallest element.
    8. Subtract the second matrix from the first.
    9. Multiply two matrices.
    10. Exit.
4. Assignments on passing pointers in functions
5. Using pointers, write your own functions for the following:
    i. String comparison
    ii. String concatenate
    iii. String copy
    iv. String length

DAY_04

DATE : 15.10.2026

TIME : 6:30 P.PM.TO 8:30 P.M.

1. Write a student class and use it in your program. Store the data of 10 students and display the sorted data according to their roll numbers, dates of birth, and total marks.
2. Implement constructor and destructors through your program.

DAY_05

DATE : 16.10.2026

TIME : 6:30 P.PM.TO 8:30 P.M.

Write a menu-driven C++ program to perform the following operations on a 3 × 3 matrix.

1. Accept matrix elements from the user.
2. Display the main diagonal elements.
3. Calculate the sum of the main diagonal elements.
4. Find the transpose of the matrix.
5. Check whether the matrix is an identity matrix.
6. Check whether the matrix is a symmetric matrix.
7. Count the positive, negative, and zero elements.
8. Search for a given element in the matrix.
9. Replace a specified element at a given position.
10. Interchange the first and third rows.
11. Interchange the first and third columns.
12. Exit.

DAY_06

DATE : 17.10.2026

TIME : 6:30 P.PM.TO 8:30 P.M.

1. Design a hierarchy of computer printers. Use multiple inheritance in your hierarchy.
2. Implement Abstract classes in your program
3. Using virtual and pure virtual functions implement hierarchy of computer printers
4. Implement diamond problem with real life example
5. Write Date and Time classes that allow you to add, subtract, read and print simple dates in dd/mm/yyyy and time in hh:mm:ss formats. Use function overloading in your program.

DAY_07

DATE : 19.10.2026

TIME : 6:30 P.PM.TO 8:30 P.M.

1. Implement exceptions and do proper management through your program.
2. Implement Custom exception class.
3. Assignments on files doing different operations

DAY_08

DATE : 21.10.2026

TIME : 6:30 P.PM.TO 8:30 P.M.

1. Assignments on templates and STL Library (stack , vector , queue , map).
2. Implement friend functions and classes in your program.
3. Implement Inline functions in your program

DAY_09

Assignment 1: Student Marks Calculator

Duration: 2 hours
Topics: Variables, operators, conditional statements, loops, arrays, and functions.
Problem Statement: Develop a C++17 program to manage student marks and generate a performance report.

Requirements:

1. Accept student details and marks in five subjects.
2. Calculate total marks, percentage, and grade using functions.
3. Determine pass or fail based on a minimum passing mark of 40 in each subject.
4. Store multiple students using arrays and display their results.
5. Sort students by percentage in descending order.

DAY_10

Assignment 2: String Operations Using Pointers

Duration: 2 hours
Topics: Pointers, arrays, functions, dynamic memory allocation, and `new/delete`.
Problem Statement: Develop a C++ program to perform basic string operations using character pointers without using the equivalent standard string-manipulation functions.

Requirements:

1. Implement custom functions for string length, copying, concatenation, and comparison.
2. Pass character pointers to functions.
3. Demonstrate dynamic memory allocation and deallocation.
4. Handle empty strings and ensure allocated memory is released correctly.

DAY_11

Assignment 3: Library Automation System

Duration: 2 hours
Topics: Classes, objects, constructors, destructors, encapsulation, static members, and operator overloading.
Problem Statement: Design a simple Library Management System using object-oriented programming.

Requirements:

1. Create a `Book` class with book ID, title, author, and availability status.
2. Implement constructors and a destructor.
3. Provide functions to add, search, issue, return, and display books.
4. Use a static member to track the number of book objects.
5. Overload the == operator to compare books by ID.

DAY_12

Assignment 4: Employee Payroll Using Inheritance

Duration: 2 hours
Topics: Inheritance, virtual functions, abstract classes, and runtime polymorphism.

Problem Statement: Develop an Employee Management System that calculates salaries for different categories of employees.

Requirements:

1. Create an abstract base class `Employee`.
2. Derive `PermanentEmployee` and `ContractEmployee` classes.
3. Implement a virtual function `calculateSalary()` in each derived class.
4. Store and process employees through base-class pointers.
5. Demonstrate runtime polymorphism by displaying employee details and salaries.

DAY_13

Assignment 5: Banking System with Exception Handling

Duration: 2 hours
Topics: Exception handling, `try`, `throw`, `catch`, and custom exceptions.
Problem Statement: Implement a simple Banking System that supports deposits and withdrawals with appropriate error handling.

Requirements:

1. Create a `BankAccount` class with account number and balance.
2. Implement deposit and withdrawal operations.
3. Throw an exception for insufficient funds or invalid transaction amounts.
4. Create and use a custom exception class for at least one error condition.
5. Handle exceptions and display meaningful error messages.

DAY_14

Assignment 6: File-Based Inventory Management

Duration: 2 hours
Topics: File handling, streams, formatted I/O, and file error handling.
Problem Statement: Develop an Inventory Management System that stores product records in a file and retrieves them when required.

Requirements:

1. Create a `Product` class with product ID, name, price, and quantity.
2. Write product records to a file using `ofstream`.
3. Read and display records using `ifstream`.

4. Calculate and display the total inventory value.
5. Handle file-opening failures and invalid input.

DAY_15

Assignment 7: Student Records Using Templates and STL

Duration: 2 hours
Topics: Function templates, class templates, STL containers, algorithms, and RTTI.
Problem Statement: Develop a Student Record Management System using C++ templates and the Standard Template Library.

Requirements:

1. Create a `Student` class with roll number, name, and marks.
2. Store student records in a `vector`.
3. Use a `map` to associate roll numbers with student details.
4. Use STL algorithms such as `sort` and `find_if` to sort and search records.
5. Implement a function template to find the maximum marks.
6. Demonstrate RTTI using `typeid`.
